

Complete 100 Days Python Project Series

A complete roadmap from Python basics to advanced AI/ML and research-level projects. This roadmap is designed for portfolio building, internships, placements, AI/ML careers, and PhD preparation.

Phase 1 — Python Fundamentals (Day 1–20)

Day	Project
Day 1	Calculator
Day 2	Number Guessing Game
Day 3	Password Generator
Day 4	Unit Converter
Day 5	BMI Calculator
Day 6	Simple Banking System
Day 7	Student Grade System
Day 8	To-Do List App
Day 9	Quiz Game
Day 10	Contact Book
Day 11	Rock Paper Scissors Game
Day 12	Dice Rolling Simulator
Day 13	Alarm Clock
Day 14	Countdown Timer
Day 15	Typing Speed Test
Day 16	Basic File Organizer
Day 17	Expense Splitter
Day 18	Attendance Management
Day 19	Library Management System
Day 20	Mini ATM Simulation

Phase 2 — Intermediate Python (Day 21–40)

Day	Project
Day 21	Weather App
Day 22	Currency Converter API
Day 23	News Headlines Scraper
Day 24	YouTube Video Downloader
Day 25	PDF Merger Tool
Day 26	QR Code Generator
Day 27	Image Resizer Tool
Day 28	Email Sender Automation
Day 29	CSV Data Analyzer
Day 30	File Encryption Tool
Day 31	Chat Application
Day 32	GUI Calculator
Day 33	GUI Expense Tracker
Day 34	Personal Diary App
Day 35	Web Scraper Dashboard
Day 36	URL Shortener
Day 37	Resume Parser
Day 38	Voice Assistant
Day 39	Desktop Notification App
Day 40	Password Manager

Phase 3 — Real-World Applications (Day 41–60)

Day	Project
Day 41	COVID Data Dashboard
Day 42	Stock Price Visualizer
Day 43	IPL Data Analysis

Day 44	Movie Recommendation System
Day 45	Face Detection System
Day 46	Object Detection
Day 47	Handwritten Digit Recognition
Day 48	Fake News Detection
Day 49	Spam Email Classifier
Day 50	Sentiment Analysis System
Day 51	House Price Prediction
Day 52	Student Performance Prediction
Day 53	Heart Disease Prediction
Day 54	Loan Approval Prediction
Day 55	Customer Churn Prediction
Day 56	Sales Forecasting
Day 57	Traffic Sign Recognition
Day 58	OCR Text Extraction
Day 59	Face Mask Detection
Day 60	Human Pose Estimation

Phase 4 — Machine Learning & Deep Learning (Day 61–80)

Day	Project
Day 61	CNN Image Classifier
Day 62	Cat vs Dog Classifier
Day 63	Transfer Learning Project
Day 64	Time Series Forecasting
Day 65	LSTM Weather Prediction
Day 66	NLP Chatbot
Day 67	Resume Ranking System
Day 68	AI Interview Bot

Day 69	Recommendation Engine
Day 70	Speech Emotion Recognition
Day 71	Sign Language Detection
Day 72	Neural Style Transfer
Day 73	Deepfake Detection
Day 74	Video Classification System
Day 75	Real-Time Crowd Detection
Day 76	AI Fitness Trainer
Day 77	ECG Signal Classification
Day 78	Shadow Detection in Videos
Day 79	Financial Time-Series Prediction
Day 80	Geometry-Aware ML Prototype

Phase 5 — Advanced Resume/Research Projects (Day 81–100)

Day	Project
Day 81	AI Healthcare Dashboard
Day 82	ECG Signal Web Application
Day 83	Medical Image Segmentation
Day 84	Smart Surveillance System
Day 85	Real-Time Object Tracking
Day 86	Human Activity Recognition
Day 87	AI-Based Attendance System
Day 88	Portfolio Optimization System
Day 89	Volatility Prediction System
Day 90	Financial Risk Modeling
Day 91	Explainable AI Dashboard
Day 92	Transformer-based Prediction System
Day 93	Geometry-Aware Recommender System

Day 94	Graph Neural Network Project
Day 95	SPD Manifold Learning Prototype
Day 96	Research Paper Reproduction
Day 97	Streamlit AI Deployment
Day 98	Docker Deployment Project
Day 99	Full Portfolio Website
Day 100	Final Capstone AI Project

Outcome After Completing This 100-Day Journey:

- Strong Python programming skills
- Real-world project experience
- AI/ML & Deep Learning portfolio
- Research-oriented projects for PhD preparation
- GitHub-ready professional portfolio
- Resume-ready advanced systems